

Introduction to Number Theory

Division

- If a and b are integers with $a \neq 0$, we say that a *divides* b if there is an integer c such that $b = ac$ (or equivalently, if b/a is an integer). When a divides b we say that a is a *factor* or *divisor* of b , and that b is a *multiple* of a . The notation $a \mid b$ denotes that a divides b . We write $a \nmid b$ when a does not divide b .
- We can express $a \mid b$ using quantifiers as $\exists c (ac = b)$, where the universe of discourse is the set of integers.
- Example : Determine whether $3 \mid 7$ and whether $3 \mid 12$.
We see that $3 \nmid 7$, because $7/3$ is not an integer. On the other hand, $3 \mid 12$ because $12/3 = 4$.

The Division Algorithm

- Theorem : Let a be an integer and d a positive integer. Then there are unique integers q and r , with $0 \leq r < d$, such that $a = qd + r$.
- **Terminology:** a – dividend d – divisor q – quotient r – remainder.
Also $q = a \text{ *div* } d$ and $r = a \text{ *mod* } d$.
- **Note the remainder (r) can not be a negative value.** The integer a is divisible by the integer d if and only if the remainder is zero when a is divided by d (ie $a = qd$).

When a divides b we say that a is a **factor** or **divisor** of b , and that b is a multiple of a and is denoted by $a \mid b$.

Modular arithmetic

Division of an integer by a positive integer produces a **quotient** and a **remainder**. Working with these remainders leads to **modular arithmetic**, which plays an important role in mathematics and computer science.

Number theory studies the properties of integers $Z = \{0, \pm 1, \pm 2, \dots\}$ and rational numbers $Q = \{p/q : p, q \in Z, q \neq 0\}$

congruence

- Definition: If a and b are integers and m is a positive integer, then a is congruent to b modulo m if m divides $a - b$.
- We use the notation $a \equiv b \pmod{m}$ to indicate that **a is congruent to b modulo m** .
- We say that $a \equiv b \pmod{m}$ is a **congruence** and that m is its **modulus** (plural **moduli**).
- If a and b are **not congruent** modulo m , we write $a \not\equiv b \pmod{m}$.

Example

- What are the quotient ($101 \text{ div } 11$) and remainder ($101 \text{ mod } 11$) when 101 is divided by 11?

- We have $101 = 11 \cdot 9 + 2$.

Hence, the quotient when 101 is divided by 11 is 9 ($101 \text{ div } 11$),

The remainder ($101 \text{ mod } 11$) is 2

The notation used to express the quotient (q) and the remainder(r) when $a = dq + r$:

$$q = a \text{ div } d,$$

$$r = a \text{ mod } d$$

Example

- What are the quotient and remainder when -11 is divided by 3 ?
- $-11 = 3(-4) + 1$ (not $-11 = 3(-3) - 2$)

Quotient = $-11 \text{ div } 3 = -4$

Remainder = $-11 \text{ mod } 3 = 1$

Theorem

- Let a and b be integers, and let m be a positive integer.

Then $a \equiv b \pmod{m}$ if and only if $a \bmod m = b \bmod m$.

$a \bmod m$ and $b \bmod m$ are the remainders when a and b divided by m

Let $a \equiv b \pmod{m}$ then $a - b = k m$ for some integer k .

Let $a = q_1 m + r_1$ and $b = q_2 m + r_2$ where r_1 and r_2 are less than m .

Since $a - b = k m$ $q_1 m + r_1 - (q_2 m + r_2) = k m$

$$r_1 - r_2 = m (k - q_1 + q_2)$$

$r_1 - r_2$ is a multiple of m and both r_1 and r_2 are less than m

$$r_1 - r_2 = 0 \times m = 0$$

$$r_1 = r_2$$

that is $a \bmod m = b \bmod m$.

Theorem

Now Let $a \bmod m = b \bmod m = r$

Then $a = q_1m + r$ and $b = q_2m + r$

$$a - b = m (q_1 - q_2)$$

$a - b$ is a multiple of m or m divides $a-b$.

That is $a \equiv b \pmod{m}$

Basic Facts About Division

Theorem 1: Let a , b , and c be integers, where $a \neq 0$. Then

1. If $a|b$ and $a|c$ then $a|(b + c)$;
2. If $a|b$ then $a|bx$ for all integers x ;
3. If $a|b$ and $b|c$ then $a|c$.

Suppose that $a | b$ and $a | c$. Then, there are integers s and t with $b = as$ and $c = at$.

Hence, $b + c = as + at = a(s + t)$.

Therefore, $b + c$ is divisible by a .

Similarly part 2 and 3 can be proved.

Basic Facts About Division

Corollary 2: If a, b , and c are integers, where $a \neq 0$, such that $a|b$ and $a|c$, then $a|mb + nc$ whenever m and n are integers.

By the definition of divisibility, if $a | b$, there exists an integer k_1 such that:

$$b = a k_1$$

Similarly, since $a | c$, there exists an integer k_2 such that:

$$c = a k_2$$

$$\begin{aligned} \text{Thus } mb + nc &= m(a k_1) + n(a k_2) \\ &= a(m k_1 + n k_2) \\ &= a k_3 \end{aligned}$$

That is $a | mb + nc$

Modular Arithmetic

If a is congruent to b modulo m ($a \equiv b \pmod{m}$) then b is congruent to a modulo m ($b \equiv a \pmod{m}$).

Example: Determine whether 17 is congruent to 5 modulo 6.

$17 \equiv 5 \pmod{6}$?

$17 - 5 = 12 = 6 * 2$ hence 17 is congruent to 5 modulo 6.

Some Basic Results

Definition: If $a \equiv b \pmod{m}$, then b is a **residue** of a modulo m . When $0 \leq b < m$, then b is called the **least non-negative residue** of a modulo m .

Theorem 5: Let m be a positive integer. The integers a and b are congruent modulo m ($a \equiv b \pmod{m}$) if and only if there is an integer k such that $a = b + km$.

Proof:

Suppose ($a \equiv b \pmod{m}$)
 $a - b = km$ for some integer k ,
 $A = b + km$

Proof:

Now Suppose $a = b + km$ for some integer k
 $a - b = km$
ie $a \equiv b \pmod{m}$

Some Basic Results

Theorem 6: Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

1. $a + c \equiv b + d \pmod{m}$
2. $ac \equiv bd \pmod{m}$
3. $a^n \equiv b^n \pmod{m}$ for any $n \geq 1$.

Example

Note: For any $a \in \mathbb{Z}$, and $m \in \mathbb{Z}^+$, we have $a \equiv (a \bmod m) \pmod{m}$

$a = m * q + r$ for some integers q and r

But $r = a \bmod m$

$a = m * q + a \bmod m$

$a - a \bmod m = m * q$

$a - a \bmod m$ is divisible by q

$a \equiv (a \bmod m) \pmod{m}$

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Some Applications of Theorem 6

$$4^2 \equiv 3 \pmod{13};$$

$$4^4 \equiv 3^2 = 9 \equiv -4 \pmod{13};$$

$$4^8 \equiv (-4)^2 \equiv 16 \equiv 3 \pmod{13};$$

$$4^{16} \equiv 3^2 \equiv 9 \equiv -4 \pmod{13};$$

$$4^{32} \equiv (-4)^2 \equiv 16 \equiv 3 \pmod{13};$$

$$4^{64} \equiv 3^2 \equiv 9 \equiv -4 \pmod{13};$$

Example

Prove that 2^{2^5} is divisible by 641.

Solution: $641 = 640 + 1 = 5 \times 2^7 + 1$

$$\Rightarrow 5 \times 2^7 \equiv -1 \pmod{641}$$

$$\Rightarrow (5 \times 2^7)^4 \equiv (-1)^4 \pmod{641}$$

$$\Rightarrow 5^4 \times 2^{28} \equiv 1 \pmod{641}$$

$$\Rightarrow (625 \pmod{641}) \times 2^{28} \equiv 1 \pmod{641}$$

$$\Rightarrow -2^4 \times 2^{28} \equiv 1 \pmod{641}$$

$$\Rightarrow 2^{32} \equiv -1 \pmod{641}$$

$$\Rightarrow 2^{2^5} + 1 \text{ is divisible by } 641.$$

Arithmetic on $\mathbb{Z}_m$

Let $\mathbb{Z}_m = \{0, 1, 2, \dots, m-1\}$, where m is a positive integer. We can define two arithmetic operations $+_m$ and $\times_m$ on $\mathbb{Z}_m$ by

$$a +_m b = (a + b) \bmod m$$

$$a \times_m b = (a \times b) \bmod m \quad \text{for all } a, b \in \mathbb{Z}_m.$$

The operations $+_m$ and $\times_m$ are called addition and multiplication modulo m and when we use these operations, we are said to be doing arithmetic modulo m .

Example: Use the definition of addition and multiplication in $\mathbb{Z}_{11}$ to find $8+_ {11} 9$ and $7\times_{11} 9$.

Primes

Definition: An integer p greater than 1 is called **prime** if the only positive factors of p are 1 and p . A positive integer that is greater than 1 and is not prime is called **composite**.

THE FUNDAMENTAL THEOREM OF ARITHMETIC: Every integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of non-decreasing size.

The Set of all Primes is Infinite

Theorem 8: There are infinitely many primes.

Proof: We assume that there are only finitely many primes, $p_1, p_2, \dots, p_n$.

Let $q = p_1 p_2 \cdots p_n + 1$.

By the fundamental theorem of arithmetic, q is prime or else it can be written as the product of two or more primes.

If p_j divides q for some j , then p_j divides $q - p_1 p_2 \cdots p_n = 1$. Therefore, p_j does not divide q for every $j = 1, 2, \dots, n$.

Hence, there is a prime not in the list $p_1, p_2, \dots, p_n$. This prime is either q , if it is prime, or a prime factor of q .

This is a contradiction because we assumed that we have listed all the primes.

Consequently, there are infinitely many primes.

Greatest Common Divisor

Definition: Let a and b be integers, not both zero. The largest integer d such that $d \mid a$ and $d \mid b$ is called the greatest common divisor of a and b and is denoted by $\gcd(a, b)$.

Definition: The integers a and b are relatively prime if their greatest common divisor is 1.

Definition: The integers $a_1, a_2, \dots, a_n$ are pairwise relatively prime if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$.

Example: Integers 17 and 22 are relatively prime, because $\gcd(17, 22) = 1$.
Integers 10, 17, and 21 are pairwise relatively prime.

Least Common Multiple

Definition: The least common multiple of the positive integers a and b is the smallest positive integer that can be divided by both numbers without leaving a remainder and is denoted by $lcm(a, b)$.

Example: Find the gcd and lcm of 150 and 500.

$$150 = 3 \times 2 \times 5^2$$

$$500 = 5^3 \times 2^2$$

$$\text{gcd} = 2 \times 5^2 = 50 \text{ (lowest powers of primes)}$$

$$\text{lcm} = 2^2 \times 3 \times 5^3 = 4 \times 3 \times 125 = 1500 \text{ (highest powers of primes)}$$

$$50 \times 1500 = 150 \times 500 \text{ (see theorem 10)}$$

Theorem 10: Let a and b be positive integers. Then $ab = \text{gcd}(a, b) \cdot \text{lcm}(a, b)$.

The Euclidean Algorithm

Lemma 11: Let $a = bq + r$, where a, b, q , and r are integers. Then $\gcd(a, b) = \gcd(b, r)$.

Example: Let us compute the greatest common divisor of 945 and 2415.

$$2415 = 945 \times 2 + 525 \quad \gcd(2415, 945) = \gcd(945, 525)$$

$$945 = 525 \times 1 + 420 \quad \gcd(954, 525) = \gcd(525, 420)$$

$$525 = 420 \times 1 + 105$$

$$420 = 105 \times 4 + 0.$$

$$\therefore \gcd(2415, 945) = 105.$$

The Euclidean Algorithm

Suppose that $a, b \in \mathbb{Z}^+$ with $a \geq b$. Let $r_0 = a$ and $r_1 = b$.

Successive application of the division algorithm gives

$$r_0 = r_1 q_1 + r_2 \quad 0 \leq r_2 < r_1$$

$$r_1 = r_2 q_2 + r_3 \quad 0 \leq r_3 < r_2$$

$\vdots$

$$r_{n-2} = r_{n-1} q_{n-1} + r_n \quad 0 \leq r_n < r_{n-1}$$

$$r_{n-1} = r_n q_n .$$

Since $a = r_0 > r_1 > r_2 > \cdots \geq 0$, $r_n = 0$ for some n .

Form the above lemma, we have

$$\gcd(a, b) = \gcd(r_0, r_1) = \gcd(r_1, r_2) = \cdots = \gcd(r_{n-2}, r_{n-1}) = \gcd(r_{n-1}, r_n) = \gcd(r_n, 0) = r_n.$$

Hence, $\gcd(a, b)$ is the last nonzero remainder in the sequence of divisions.

gcds as Linear Combinations

BÉZOUT'S Theorem 12: If a and b are positive integers, then there exist integers s and t such that $\gcd(a, b) = sa + tb$.

Example: Express $\gcd(252, 198) = 18$ as a linear combination of 252 and 198.

$$252 = 1 \times 198 + 54$$

$$198 = 3 \times 54 + 36$$

$$54 = 1 \times 36 + 18$$

$$36 = 2 \cdot 18.$$

$$\therefore \gcd(252, 54) = 18.$$

$$18 = 54 - 1 \times 36$$

$$= 54 - 1(198 - 3 \times 54)$$

$$= 4 \times 54 - 1 \times 198$$

$$= 4(252 - 1 \times 198) - 1 \times 198$$

$$= 4 \times 252 - 5 \times 198.$$

$$\text{Hence } \gcd(252, 198) = \mathbf{18} = 4 \times \mathbf{252} - 5 \times \mathbf{198}.$$

Applications BÉZOUT'S Theorem

Lemma 13: If a , b , and c are positive integers such that $\gcd(a, b) = 1$ and $a|bc$, then $a|c$.

Lemma 14: If p is a prime and $p|a_1a_2 \cdots a_n$, where each a_i is an integer, then $p|a_i$ for some i .

Theorem 15: Let m be a positive integer and let a , b , and c be integers. If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then $a \equiv b \pmod{m}$.

Solving Linear Congruence

A congruence of the form $ax \equiv b \pmod{m}$ is called a linear congruence, where $a, b, m \in \mathbb{Z}$, $m > 0$, and x is a variable.

Finding all integers x that satisfy this congruence is known as solving this linear congruence.

If there exists an integer a' such that $a'a \equiv 1 \pmod{m}$, then a' is said to be an inverse of a modulo m .

Example: What are the solutions of the linear congruence $3x \equiv 4 \pmod{7}$?

Theorem 16: Let $m \geq 2$. If a and m are relatively prime then there exists a unique integer $\bar{a}$ such that $a\bar{a} \equiv 1 \pmod{m}$ and $0 < \bar{a} < m$.

Example

Find an inverse of 101 modulo 4620.

$$4620 = 45 \cdot 101 + 75$$

$$101 = 1 \cdot 75 + 26$$

$$75 = 2 \cdot 26 + 23$$

$$26 = 1 \cdot 23 + 3$$

$$23 = 7 \cdot 3 + 2$$

$$3 = 1 \cdot 2 + 1$$

$$2 = 2 \cdot 1.$$

Euler φ -function

Definition: The Euler φ -function is the function $\varphi: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ defined by $\varphi(n)$ = the number of integers in $\{1, 2, \dots, n-1\}$ which are relatively prime to n .

Proposition 17:

- If p is prime, $\varphi(p) = p - 1$.
- If p is prime and $n \geq 1$, then $\varphi(p^n) = p^n - p^{n-1}$.
- If $\gcd(m, n) = 1$, then $\varphi(mn) = \varphi(m)\varphi(n)$.
- For any integer $n > 1$, if $n = p_1^{a_1} p_2^{a_2} \cdots p_s^{a_s}$ is the prime power factorization, then $\varphi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_s}\right)$.

Example: Find $\varphi(1000)$.

Complete and Reduced residue system

Definition: For a positive integer n , a reduced residue system mod n is a set of numbers $a_1, a_2, \dots, a_{\varphi(n)}$ such that

1. If $i \neq j$ then $a_i \not\equiv a_j \pmod{n}$
2. For each i , $\gcd(a_i, n) = 1$.

Example: $\{1, 5, 7, 11\}$ is a reduced residue system mod 12. $\{-11, 17, 31, -1\}$ is also another reduced residue system for mod 12.

Definition: For a positive integer n , a complete residue system mod n is a set of numbers $\{0, 1, 2, \dots, n - 1\}$.

Basic Results

Lemma 18: Let $\varphi(n) = k$, and let $\{a_1, a_2, \dots, a_k\}$ be a reduced residue system mod n .

1. For all m , $\{a_1 + mn, a_2 + mn, \dots, a_k + mn\}$ is a reduced residue system mod n .
2. If $(m, n) = 1$, $\{ma_1, ma_2, \dots, ma_k\}$ is a reduced residue system mod n .

Corollary 19: Let $\varphi(n) = k$, and let $\{a_1, a_2, \dots, a_k\}$ be a reduced residue system mod n . Suppose $\gcd(s, n) = 1$, and let t be any integer.

Then $\{sa_1 + tn, sa_2 + tn, \dots, sa_k + tn\}$ is a reduced residue system mod n .

Euler Theorem

Theorem 20: (Euler) Let $n > 0$, $\gcd(a, n) = 1$. Then $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Proof: Let $\varphi(n) = k$, and let $\{a_1, a_2, \dots, a_k\}$ be a reduced residue system mod n . We may assume that the a_i 's lie in the range $\{1, \dots, n-1\}$.

Since $\gcd(a, n) = 1$, $\{aa_1, aa_2, \dots, aa_k\}$, is another reduced residue system mod n . Since this is the same set of numbers mod n as the original system, the two systems must have the same product mod n :

$$\begin{aligned} aa_1 \times aa_2 \times \dots \times aa_k &\equiv a_1 \times a_2 \times \dots \times a_k \pmod{n} \\ \Rightarrow a^k (a_1 \times a_2 \times \dots \times a_k) &\equiv a_1 \times a_2 \times \dots \times a_k \pmod{n} \\ \Rightarrow a^k &\equiv 1 \pmod{n} \quad (\because \gcd(a_i, n) = 1 \text{ \& applying theorem 15}). \end{aligned}$$

That is; $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Applications of Euler's Theorem

Theorem 21 (Fermat's Little Theorem): Let p be any prime number and suppose that p does not divide a . Then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Example 1: Find $9^{16} \pmod{40}$.

$\varphi(40) = 16$, and $\gcd(9, 40) = 1$. Hence by Euler's Theorem $9^{16} \equiv 1 \pmod{40}$.

Example 2: Find $7^{222} \pmod{11}$.

Example 3: Solve $15x \equiv 7 \pmod{32}$.